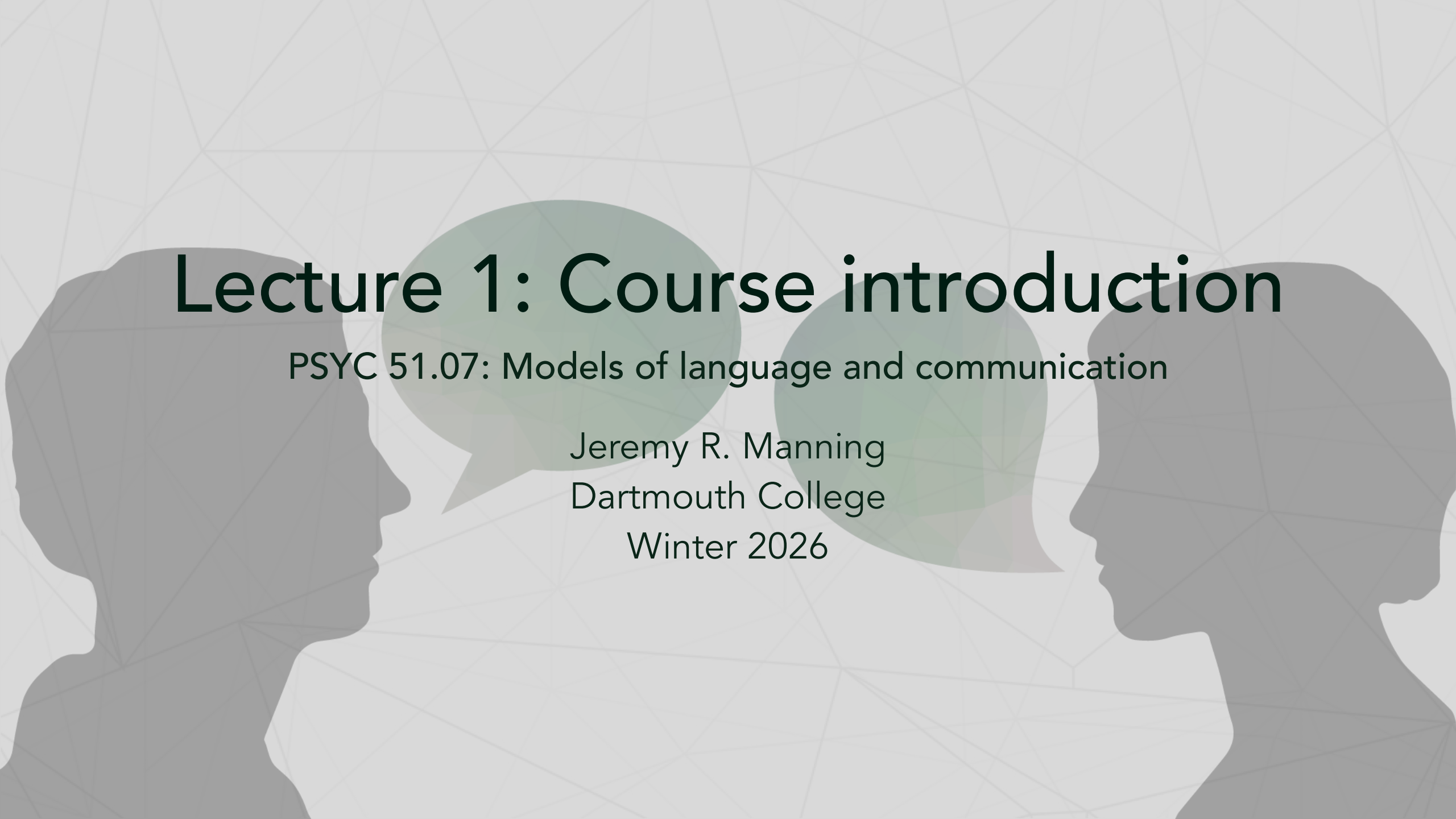




Lecture 1: Course introduction

PSYC 51.07: Models of language and communication

Jeremy R. Manning
Dartmouth College
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About the instructor

Jeremy R. Manning, Ph.D.

Associate Professor | Psychological & Brain Sciences |  | Moore 349

 context-lab.com

 ContextLab

Research focus

How do our brains support our ongoing conscious thoughts, and how (and what) do we remember?

Key areas

Learning and memory, education technology, brain network dynamics, data science, NLP

Approach

Theory, models, experiments, neuroimaging

Training



B.S., Neuroscience & Computer Science



Ph.D., Neuroscience



Postdoc, Computer Science & Neuroscience

Funding & collaborators



What is this course about?

Course content

We will explore how machines can understand and generate human language:

- Building conversational agents from scratch
- Understanding how language relates to thought
- Hands-on programming with real models
- Critical thinking about AI consciousness

Course design

This course is experiential! You will learn by doing: coding, experimenting, discussing, and researching.

Course structure



We'll build from pattern matching to transformer-based models through our 5 main projects

Grading

- Bi-weekly short projects (5): 75%
- Final project: 25%
- Can work individually or in groups
- See [syllabus](#) for additional details

Tools

- Google Colaboratory
- HuggingFace
- GitHub / Discord
- GenAI (Claude, ChatGPT, Gemini)

The big questions

1. Can machines truly *understand* language?
2. What is the relationship between **language** and **thought**?
3. Can statistical patterns capture **meaning**? How? Under which circumstances?
4. Can AI be conscious? If so, what are the implications?

Note

These are not just "fluff" questions: they are at the heart of cognitive science, philosophy, psychology, and neuroscience!

Discussion: Is ChatGPT conscious?

- What does "conscious" even mean?
- How would we test for consciousness?
- Does it matter if ChatGPT *seems* conscious?

For further consideration

What are the implications for

- Ourselves
- Other animals
- Other life forms in general (aliens? synthetic life?)
- Policy, ethics, and society more broadly

What is consciousness?

- **Phenomenal:** Subjective experience-- the "what it is like" quality of sensations and emotions
- **Access:** Information available for reasoning, reporting, and guiding voluntary behavior
- **Self-awareness:** Knowledge of one's own mental states, including recognizing oneself as a distinct entity

Example of phenomenal consciousness

The redness of red, pain, taste of coffee

Example of access consciousness

Being able to report on and use information to guide behavior

Example of self-awareness

Knowing that you are thinking, or recognizing your own emotions

For further consideration

If ChatGPT says "I feel happy," does it actually *feel* anything?

The hard problem

Humans

- Share similar biology
- Have our own conscious experiences
- Behave consistently with having experiences (e.g., of being human, living in the world, etc.)

AI

- Completely different "biology" (silicon vs. neurons)
- No shared evolutionary history
- Can produce human-like behavior without themselves being human

Why is it difficult to know if AI is conscious?

We cannot directly observe consciousness — even in other humans!

The Chinese room argument

John Searle (1980)

- Thought experiment about understanding vs. simulation.
- **The setup:** Person in room with Chinese symbols and rule book
- Does following rules = understanding? Searle argues: **No!**



Person follows rules but does not understand Chinese

Volition

- Another critical aspect of the human conscious experience is the ability to **decide** (how to act, what to think, etc.)
- Modern LLMs are trained to **respond** to other inputs (i.e., produce statistically likely sequence completions), but they cannot themselves initiate new or unexpected actions

Think about it!

LLMs are like a bellows that can only blow air if someone else is pumping it. When not in use, they are static. They can not sense the passage of time. They cease to exist between invocations.

Current scientific consensus

Survey says...

Most cognitive scientists and AI researchers agree: **Current LLMs are not conscious.**

- No sensory-motor grounding in the world
- No persistent self-model or goals
- Pattern matching $\neq$ understanding

Language and thought

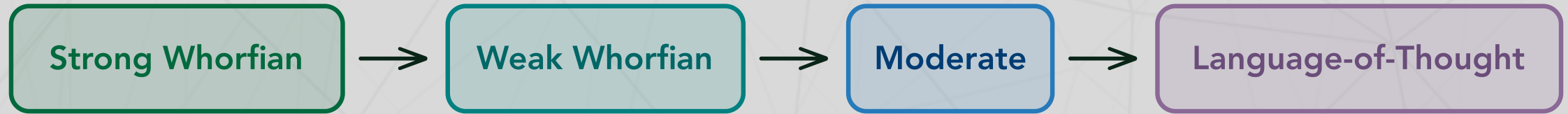
Discussion

Do you need language to think? Does language *shape* how you think?

- Possibility 1: Language is necessary for thought
- Possibility 2: Language is just a tool for communication



The language-thought spectrum



Language shapes thought (left) to language independent of thought (right)

Current evidence points toward the middle: language and thought interact in complex ways, but are not identical.

Evidence: the language network

Further reading

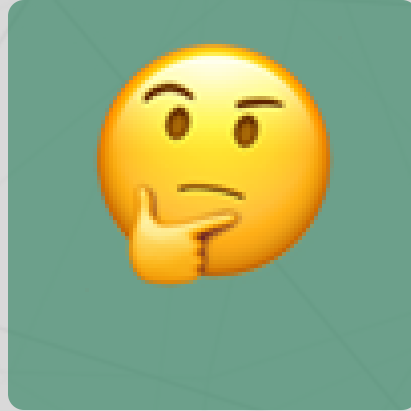
Fedorenko et al. (2024, *Nature*): The language network as a natural kind

- The brain has a **specialized language network**
- Distinct (as measured using neuroimaging and lesion studies) from: reasoning, math, social cognition, music
- **Implication:** Language and thought are *separable* in the brain!

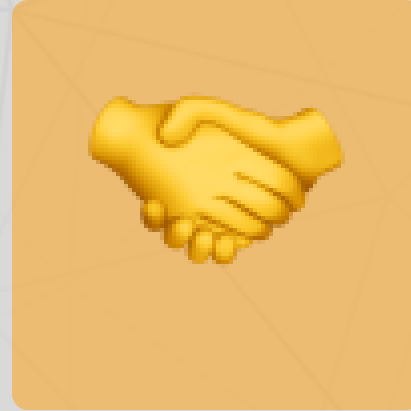
Brain networks



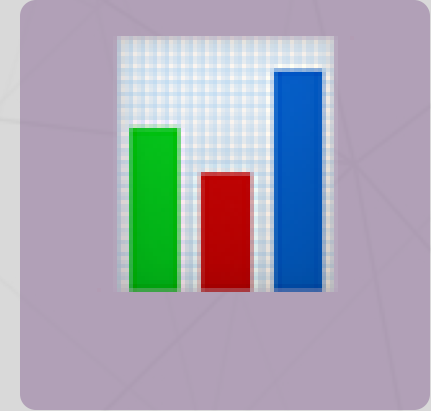
Language



Reasoning



Social



Math

Key insight

Language is a specialized system, *not* the basis of all thought! This has fascinating implications for LLMs: we've built machines that can *interact* using language, but this doesn't necessarily mean they understand it in any recognizable way.

However, language can *shape* thought

Further reading

Lupyan et al. (2020, *TiCS*): *Effects of language on visual perception*

Having words for things affects how we *see* them:

- Speed up visual search
- Alter color perception
- Influence object categorization

Example: Russian blue



English: "blue"



English: "blue"



Russian: "sinii"



Russian: "goluboy"

Key finding

English speakers are slower to distinguish shades of blue than Russian speakers: language categories affect *perception*, not just description!

So...what about LLMs?

The grounding problem

How do symbols (words) get their meaning?

- For **humans**, the meanings of symbols are learned through *experience*:
 - See, touch, taste objects
 - Act in the world
 - We learn to associate symbols with our experiences
- For **LLMs**, the meanings of symbols are learned through *statistics*:
 - Learn patterns in text
 - Generate text based on those patterns
 - No direct connection with the external world

Our approach

Note

Philosophy of this course: We will build language models *from scratch* to understand what they can and cannot do.

Critical perspective:

- Question assumptions about "understanding"
- Appreciate both capabilities and limitations
- Think about implications for cognitive science

From simple to complex



At what point does pattern matching become understanding?

Note

Discussion: At what point (if any) does pattern matching become understanding?

HuggingFace: Your learning companion

We will use HuggingFace for:

- **Course materials:** Free NLP course at huggingface.co/course
- **Pre-trained models:** Download and use state-of-the-art models
- **Datasets:** Access curated datasets for training

Warning

Assignment: Create a free HuggingFace account and complete Chapter 1 of the NLP course!

Next lectures this week

Lecture 2: Pattern matching and ELIZA

String manipulation
Regular expressions
Introduction to ELIZA

Note

Lecture 3 (X-hour): ELIZA implementation

- Complete architecture
- The ELIZA effect
- Assignment 1 details

Readings for this week

Note

Required readings:

1. Weizenbaum (1966): ELIZA
2. Fedorenko et al. (2024): The language network
3. Lupyan et al. (2020): Effects of language on visual perception

Tip: Start with Weizenbaum--it will help you understand the fundamentals!

Key takeaways

1. **Consciousness is complex:** Multiple types, hard to define
2. **Language $\neq$ Thought:** But they interact in interesting ways
3. **LLMs are not conscious:** They are sophisticated pattern matchers
4. **Grounding matters:** Meaning comes from experience
5. **We will build to understand:** Hands-on reveals true capabilities

Questions?



Email



Discord



Moore 349

Note

Office hours: By appointment | **Lab website:** context-lab.com